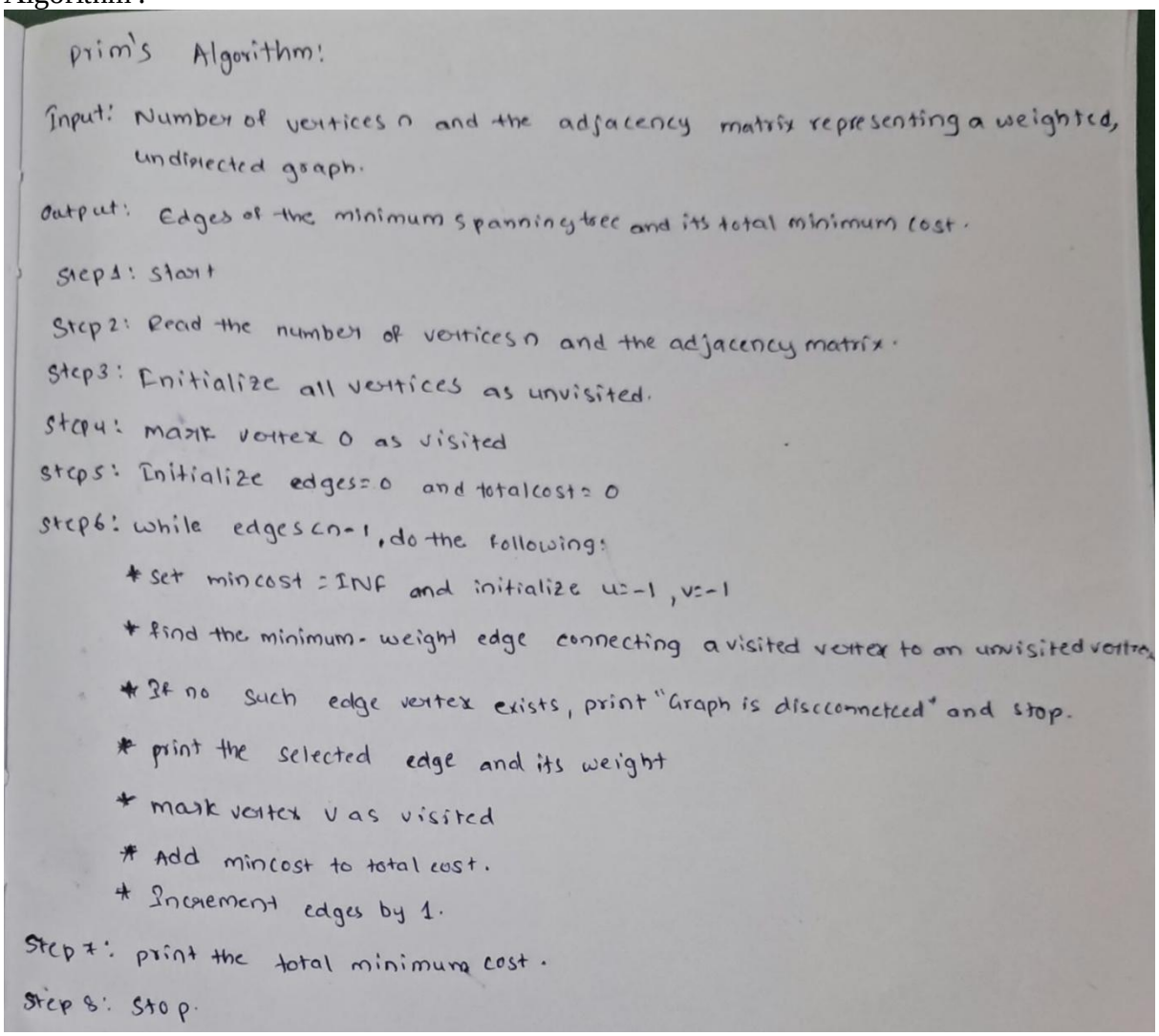
PRACTICAL - 9

AIM :Implementation of prim's algorithm.

Algorithm :



code :

```
#include <iostream>
using namespace std;
#define INF 9999
int main() {
   int n;
   cout << "Enter number of vertices: ";
   cin >> n;
```

```
    int cost[10][10];
    cout << "Enter adjacency matrix:\n";
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            cin >> cost[i][j];
            if (cost[i][j] == 0)
                cost[i][j] = INF;
        }
    }
    int visited[10] = {0};
    visited[0] = 1;
    int edges = 0, totalCost = 0;
    cout << "\nEdges in Minimum Spanning Tree:\n";
    while (edges < n - 1) {
        int minCost = INF;
        int u = -1, v = -1;
        for (int i = 0; i < n; i++) {
            if (visited[i]) {
                for (int j = 0; j < n; j++) {
                    if (!visited[j] && cost[i][j] < minCost) {
                        minCost = cost[i][j];
                        u = i;
                        v = j;
                    }
                }
            }
        }
            if (u == -1) {
            cout << "Graph is disconnected!\n";
            return 0;
        }
        cout << u << " - " << v
             << " : " << minCost << endl;

        visited[v] = 1;
        totalCost += minCost;
        edges++;
    }
    cout << "Minimum Cost = " << totalCost << endl;

    return 0;
```

}

Output :

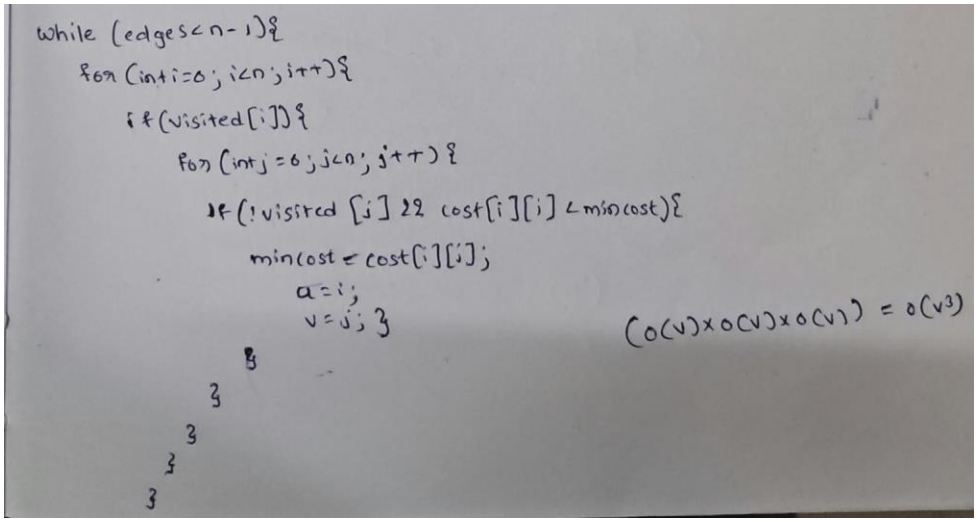
```
Output

Enter number of vertices: 4
Enter adjacency matrix:
0 2 0 6
2 0 3 8
0 3 0 1
6 8 1 0

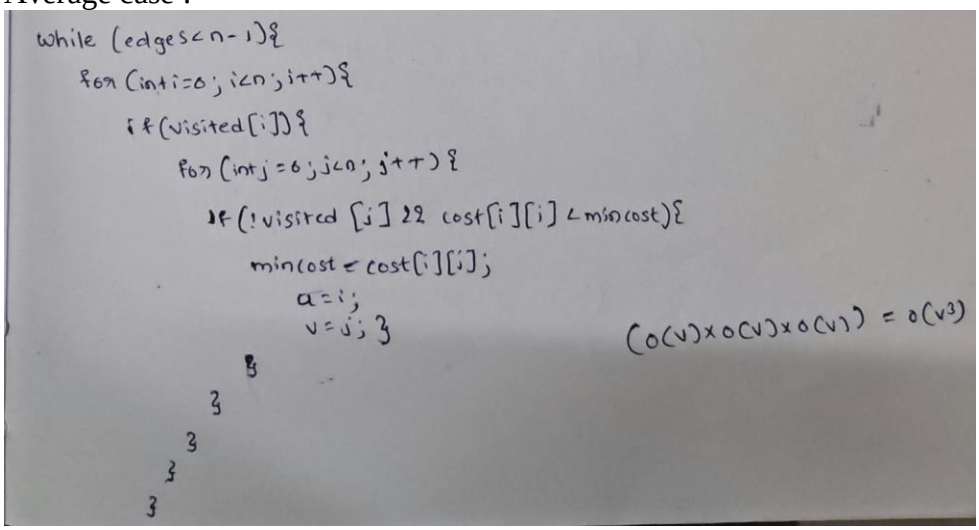
Edges in Minimum Spanning Tree:
0 - 1 : 2
1 - 2 : 3
2 - 3 : 1
Minimum Cost = 6
```

Time complexity :

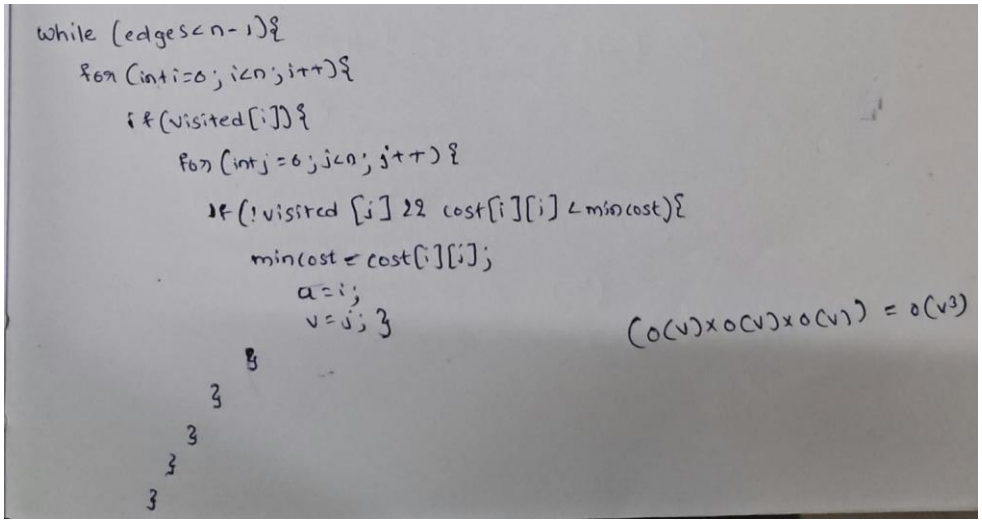
Best case :



Average case :



Worst case :



Conclusion:

Prim's Algorithm is a greedy algorithm used to find the Minimum Spanning Tree (MST) of a connected, weighted, undirected graph. It starts from any vertex and repeatedly selects the minimum-weight edge connecting a visited vertex to an unvisited vertex. The process continues until all vertices are included in the spanning tree. It helps minimize the total cost of connecting all vertices and is useful in network design, road construction, and communication systems.